SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	**Cloud Computing and Big Data Analytics**	**COURSE CODE:**	**CSA1501**

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Total Marks: 40

Duration: 60 Minutes

Code of Conduct

1 I **Ch.Abhilash Reddy(192525462)__________** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: **___Abhilash reddy_______________________**

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem -** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across

multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem -** A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
5. **Agriculture Smart Farming Problem -** A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

1. Retail Data Optimization Problem

A unified retail architecture should combine batch and streaming data so that stock levels, demand forecasts, and customer recommendations are updated continuously.

Proposed Big Data Architecture / Strategy

Data Sources: POS billing systems, e-commerce orders, warehouse inventory, product catalogues, mobile-app clicks, loyalty data and customer browsing events.

Ingestion Layer: Apache Kafka can collect high-volume events from stores and the website. APIs and Apache NiFi can ingest master data and scheduled files from ERP/CRM systems.

Storage Layer: Use a cloud data lake such as Amazon S3/Azure Data Lake/HDFS for raw and historical data. A NoSQL database such as Cassandra or DynamoDB can serve rapidly changing inventory. A cloud data warehouse such as BigQuery, Redshift or Snowflake supports BI analysis.

Processing Layer: Apache Spark performs large-scale batch processing and model training. Spark Structured Streaming or Apache Flink processes sales and stock events in near real time.

Analytics Models: Demand forecasting can use ARIMA/Prophet for time series and XGBoost/LSTM for complex seasonal patterns. Recommendations can combine collaborative filtering, content-based filtering and customer segmentation using K-means.

Serving & Visualization: Inventory alerts and recommendation APIs can be exposed through microservices. Power BI/Tableau dashboards show stock, sales, demand trends and store-level KPIs.

Working Flow

1. A sale is published to Kafka immediately.
2. The stream processor updates available stock and detects low-stock conditions.
3. Historical sales, promotions and seasonality are used to forecast future demand.
4. Recommendation models rank products for each customer based on behavior and similarity.
5. Dashboards and APIs deliver results to managers, websites and mobile applications.

Key Benefits

- Real-time stock visibility and fewer stock-outs.
- Better demand planning and replenishment.
- Personalized offers improve customer engagement.
- Scales across stores, channels and growing data volumes.

Conclusion

This architecture is scalable because storage and compute can grow independently. Streaming provides fast inventory visibility, while historical analytics improves forecasting and personalization.

2. Social Media Fraud Detection Problem

The platform requires distributed ingestion, stream analytics and machine learning because billions of interactions must be evaluated with very low delay.

Proposed Big Data Architecture / Strategy

Data Ingestion: Kafka or Pulsar partitions posts, comments, likes, shares, login events and account activity across many brokers for horizontal scalability.

Real-Time Processing: Apache Flink, Spark Structured Streaming or Kafka Streams computes event rates, repeated content, unusual account activity and propagation patterns.

Storage: HDFS/S3 stores long-term raw data; Cassandra/HBase stores high-throughput user features; Elasticsearch/OpenSearch supports fast text and investigation searches.

Machine Learning: Fake-account detection can use Random Forest, XGBoost or neural networks on profile age, posting frequency, device/IP patterns and social graph features. Graph algorithms detect coordinated account clusters. NLP/transformer models classify misleading or suspicious text.

Model Serving: Models are deployed as scalable APIs or embedded into stream jobs. Each event receives a risk score; high-risk content can be queued for review or temporarily limited.

Monitoring: Track precision, recall, F1-score, false-positive rate, latency and concept drift. Retrain models when attacker behavior changes.

Working Flow

6. User activity enters Kafka partitions.
7. Streaming jobs create real-time behavioral and graph features.
8. ML models generate account/content risk scores.
9. Rules and thresholds combine model scores with trusted signals.
10. Suspicious cases are sent to moderation, while confirmed outcomes become labels for retraining.

Key Benefits

- Fast detection of coordinated and automated abuse.
- Horizontal scalability for billions of events.
- Continuous learning from moderation feedback.
- Near real-time protection with measurable model quality.

Conclusion

Machine learning finds complex fraud patterns that fixed rules miss, while stream frameworks allow these models and features to operate continuously at large scale.

3. Government Data Platform Problem

A national platform needs governance as strongly as technology. The strategy should make data discoverable and interoperable while enforcing privacy, accountability and controlled access.

Proposed Big Data Architecture / Strategy

Governance Structure: Create a central data governance council with data owners and stewards in each department. Define responsibility for quality, classification, access approval, retention and incident response.

Common Standards: Adopt shared metadata, data dictionaries, identifiers, schemas, APIs and open formats. Geospatial datasets should use recognized GIS standards and consistent coordinate/reference metadata.

Data Catalog & Lineage: Use a catalog such as Apache Atlas, Microsoft Purview or AWS Glue Data Catalog to record dataset owners, definitions, sensitivity, lineage and usage.

Data Quality: Apply validation rules for completeness, accuracy, consistency, timeliness and duplicate detection. Publish quality scores and maintain master/reference data.

Privacy & Security: Classify data by sensitivity; encrypt data in transit and at rest; use IAM, RBAC/ABAC, MFA, network segmentation and key management. Apply masking, tokenization, pseudonymization or aggregation to personal data.

Audit & Compliance: Maintain immutable access logs, periodic access reviews, retention schedules, consent/legal-purpose records and security audits. Use least privilege and zero-trust principles.

Working Flow

11. Departments publish approved datasets through standardized ingestion pipelines.
12. Metadata and lineage are registered in the central catalog.
13. Quality and privacy controls are applied before data becomes shareable.
14. Authorized agencies discover data through the catalog and access it using governed APIs or secure workspaces.
15. Usage is logged and reviewed continuously.

Key Benefits

- Consistent data definitions across agencies.
- Secure sharing with clear ownership and accountability.
- Improved data quality, lineage and discoverability.
- Privacy-by-design and auditable access.

Conclusion

The result is a shared data ecosystem in which agencies can reuse trusted datasets without losing control over privacy, ownership, security or regulatory obligations.

4. Banking Fraud Detection Problem

The bank should use a lambda- or lakehouse-style pipeline that combines historical model training with low-latency streaming decisions.

Proposed Big Data Architecture / Strategy

Transaction Ingestion: Kafka captures card, UPI, ATM, internet-banking and account events. Partitioning by account/card allows parallel processing while preserving useful event order.

Streaming Features: Flink or Spark Structured Streaming calculates transaction velocity, amount deviation, location/device changes, failed attempts and short-window aggregates.

Historical Store: A secure data lake/lakehouse stores transactions, chargebacks, customer profiles and labels. Spark processes historical data and builds training datasets.

Algorithms: Use Logistic Regression as a baseline; Random Forest/XGBoost for nonlinear fraud patterns; Isolation Forest/Autoencoders for anomalies; graph analytics for mule accounts and suspicious networks. Combine models with expert rules.

Real-Time Scoring: A feature store provides consistent features to the deployed model. The stream job calls the model and produces a fraud probability in milliseconds.

Action Layer: Low-risk transactions pass; medium-risk transactions can trigger step-up authentication; high-risk transactions are blocked or sent to analysts. Outcomes feed back into training data.

Working Flow

16. Transaction event arrives in Kafka.
17. Streaming engine enriches it with customer, device and recent-behavior features.
18. Fraud model and rules generate a risk score.
19. Decision service approves, challenges or blocks the transaction.
20. Confirmed fraud/legitimate labels are stored for continuous retraining and threshold tuning.

Key Benefits

- Millisecond-to-second fraud decisions.
- Combined historical and streaming context.
- Lower false positives through hybrid models and feedback.
- Scalable processing for very high transaction rates.

Conclusion

High accuracy comes from combining supervised, anomaly and graph methods; low latency comes from distributed streaming, precomputed features and scalable model serving.

5. Agriculture Smart Farming Problem

Precision agriculture can use an IoT-to-cloud big data platform that combines sensor streams, drone imagery, weather data and historical farm records.

Proposed Big Data Architecture / Strategy

Data Sources: Soil moisture, pH, temperature, humidity, nutrient sensors, weather stations, GPS-enabled equipment, drone/multispectral images and crop records.

Edge & Ingestion: Gateways perform basic filtering and buffering in the field. MQTT/IoT Hub sends telemetry to Kafka or a cloud event service. Intermittent connectivity can be handled by store-and-forward at the edge.

Storage: A time-series database stores sensor measurements; object storage/data lake stores drone images and historical datasets; geospatial databases such as PostGIS support field-level location analysis.

Processing: Spark processes historical and image-derived features; Flink/Spark Streaming evaluates sensor events in real time. GIS tools combine soil, weather and field boundaries.

Analytics: Regression/XGBoost predicts yield; time-series models forecast moisture/weather effects; classification detects crop stress or disease; computer vision analyzes drone images; optimization models recommend irrigation and fertilizer quantities.

Decision Support: Dashboards and mobile alerts show moisture maps, disease risk, irrigation schedules, expected yield and abnormal sensor conditions.

Working Flow

21. Sensors and drones collect field data.
22. Edge devices clean and transmit observations.
23. Cloud pipelines integrate sensor, weather and geospatial data.
24. Analytics models estimate crop condition, yield and resource needs.
25. Farmers receive field-specific recommendations and alerts; actual outcomes improve future models.

Key Benefits

- Reduced water, fertilizer and energy waste.
- Earlier disease/stress detection.
- Location-specific recommendations instead of uniform treatment.
- Improved yield forecasting and farm productivity.

Conclusion

Analytics converts distributed farm data into practical actions, helping farmers use water and fertilizer more efficiently, detect problems earlier and improve yield.